



Krypton

K Y P

A fast, lightweight and fair
Proof-of-Work blockchain.

Whitepaper

Version 2.0 — Protocol v1.6

Mainnet launched on 24 July 2025

No premine · No private funding · 100% community-driven

Whitepaper

1. Abstract
2. Vision and founding principles
3. Technical specifications
4. Monetary emission
5. Consensus and security
6. Peer-to-peer network
7. Mining
8. Wallet and fees
9. Governance
10. Use cases
11. Roadmap
12. Resources and disclaimer

1. Abstract

Krypton (KYP) is an open-source cryptocurrency derived from Litecoin, itself derived from Bitcoin. It uses Script proof-of-work, targets one block every 60 seconds, and distributes its entire supply through mining: no premine, no founder allocation, no venture capital.

The project aims for a chain that is simple, understandable and durable rather than feature-heavy. The code is public, auditable, and inherits more than a decade of hardening from the Bitcoin protocol. Protocol version 1.6, described in this document, makes difficulty adjustment more responsive and introduces explicit protection against deep chain reorganizations.

60 s

BLOCK TIME

15 M

MAX KYP SUPPLY

0 %

PREMINE

2. Vision and founding principles

Krypton starts from a simple observation: a digital currency does not need to be complex to be useful. It needs to be predictable, verifiable by anyone, and beyond the control of any single entity.

- **Fair start.** The first block was mined under the same conditions as the latest one. Nobody holds a structural head start over the rest of the network.

- **Deliberate simplicity.** No staking, no masternodes, no tokenized governance. One security rule only: proof of work.
- **Full transparency.** Every line of code, every consensus parameter and every change is public on GitHub and discussed in the open.
- **Durability.** Technical decisions favour long-term stability over announcement effects.

3. Technical specifications

Parameter	Value
Name / Symbol	Krypton / KYP
Algorithm	Scrypt (Proof-of-Work)
Target block time	60 seconds
Initial block reward	14.26940639 KYP
Halving	50% every 525,600 blocks (~1 year)
Maximum supply	14,999,999.9078 KYP
Distribution period	~31 years (reward reaches 0 at block 16,293,600)
Difficulty adjustment	LWMA — 120 blocks, then 60 blocks from block 527,040
Maximum reorg depth	200 blocks (~3h20), configurable
Addresses	K... (legacy), A... (P2SH), kyp1... (bech32)
P2P / RPC port	8369 / 8370 (testnet: 18369 / 18370)
Premine	None
Licence	MIT

4. Monetary emission

The block reward halves every 525,600 blocks, roughly once a year. Emission therefore decays sharply: half of the total supply is created during the first year, and 99% within about seven years.

Year	Reward per block	Total mined	Share of supply
1	14.26940639	7,500,000	50.0%
2	7.13470319	11,250,000	75.0%
3	3.56735159	13,125,000	87.5%
4	1.78367579	14,062,500	93.8%

7	0.11147973	14,941,406	99.6%
~31	0	14,999,999.9078	100%

The final total falls marginally short of 15,000,000 KYP. This is not an approximation but an arithmetic consequence: each halving is an integer right-shift, truncating a fraction of a satoshi every era. Bitcoin exhibits exactly the same behaviour.

Design decision: since version 1.6 the reward is computed from an exact integer constant (1,426,940,639 satoshis) rather than a floating-point literal. The emitted value is strictly identical, but the result is now guaranteed bit-for-bit across every platform and compiler. Floating-point arithmetic in consensus code is a classic source of accidental chain splits.

5. Consensus and security

5.1 Script proof of work

Krypton uses Script, a memory-hard key derivation function. Compared with Bitcoin's SHA-256, it reduces the relative advantage of specialised hardware and keeps CPU and GPU mining viable for longer.

5.2 LWMA difficulty adjustment

Difficulty is recalculated at every block using a Linearly Weighted Moving Average, which gives more weight to recent blocks. Unlike Bitcoin's epoch-based retarget, it reacts continuously and copes well with abrupt swings in hashrate.

From block **527,040**, the observation window narrows from 120 to 60 blocks. Difficulty then tracks hashrate changes roughly twice as fast, shortening the periods of unusually slow blocks after a large miner leaves.

Warning — consensus fork. The window change at block 527,040 is a *hard fork*. Any node or mining pool not running version 1.6 or later before that height will be stranded on a divergent chain. The upgrade must be coordinated network-wide.

5.3 Deep reorganization protection

A so-called "51% attack" consists of secretly building a longer competing chain and then publishing it to reverse already-confirmed transactions. Since version 1.6, a Krypton node refuses any alternative chain that would reorganize more than **200 blocks** (about 3h20) below its current tip, even when that chain carries more cumulative work.

The implementation is deliberately conservative:

- **Header-level rejection.** The offending chain is discarded before its blocks are ever downloaded, so it can neither consume bandwidth nor stall the node's own legitimate chain.
- **Inactive during initial sync.** A node downloading the chain for the first time is never affected.
- **No peer punishment.** An honest node sitting on a legitimate old fork has its header refused but is not banned.

- **Configurable.** The `-maxreorgdepth=<n>` option adjusts or fully disables the mechanism, which remains local node policy rather than a strict consensus rule.

This protection does not replace the security provided by hashing power: it bounds the damage a successful attack could inflict on confirmed history.

6. Peer-to-peer network

Krypton inherits Bitcoin Core's networking layer, whose timeouts were nonetheless calibrated for ten-minute blocks. Version 1.6 recomputes them from the chain's real interval: stale-tip detection, extra-peer eviction and chain-sync timeout. The network therefore reacts about ten times faster to a stalled or lagging peer, improving block propagation on a 60-second chain.

7. Mining

Mining is open to everyone, with no permission or registration. Script allows CPU and GPU mining; dedicated Script hardware also exists. There is no staking, no masternodes and no deferred reward mechanism: 100% of emission goes through proof of work.

A full node is enough to mine solo. Joining a pool requires Script-compatible mining software pointed at the pool's address.

8. Wallet and fees

The official wallet ships as a graphical application (`krypton-qt`) and as command-line tools (`kryptond` , `krypton-cli`) for Windows, macOS and Linux. Keys are held locally: no third-party server holds user funds.

The fee structure is inherited from Litecoin and was reviewed for version 1.6. Fees remain deliberately low and proportional to transaction size rather than amount. On a chain producing a block every 60 seconds, the fee estimator collects ten times more data points per hour than on Bitcoin, making its predictions more responsive.

Back up your wallet. The `wallet.dat` file holds your private keys. Without it and without a backup, the corresponding funds are permanently lost: no recovery procedure exists on a blockchain.

9. Governance

Krypton has no foundation, no company and no board. Protocol changes are proposed and debated publicly on GitHub, then adopted by node operators and miners who choose — or decline — to run the new version. No entity can unilaterally change the rules: a consensus change the network does not accept simply has no effect.

10. Use cases

- **Fast payments.** One-minute average confirmation, against ten for Bitcoin.
- **Borderless transfers.** No intermediary, no prior authorisation, fees independent of distance and amount.
- **Micropayments.** Low fees and divisibility down to one hundred-millionth of a unit.

- **Programmed store of value.** A capped, public emission verifiable by anyone running a node.

11. Roadmap

The roadmap below is ordered by dependency rather than by date. Each stage unlocks the next: a mobile wallet needs lightweight-client infrastructure, hardware-wallet and exchange integrations need that same infrastructure, and every one of them needs a healthy, well distributed network underneath. No delivery dates are given — items ship when they are ready and properly tested, not to meet an announcement.

Delivered

Milestone	Status
Genesis block and mainnet launch	Completed
Public block explorer	Live
Windows, macOS and Linux wallets (GUI and command line)	Completed
Protocol v1.6 — responsive difficulty, deep-reorg protection, P2P tuning	Released

Stage 1 — Network foundations

Nothing else can be built on a fragile base. This stage is about making the chain robust and easy to join.

Objective	Status
Difficulty fork activation and network-wide upgrade coordination	Scheduled
Redundant seed nodes and DNS seeds	Planned
Mining pool support, so miners are not limited to solo mining	Planned
User and miner documentation	In progress
Network dashboard (nodes, hashrate, difficulty)	In progress

Stage 2 — Lightweight client infrastructure

A phone cannot download the whole blockchain, and neither can most third-party services. An indexing server (Electrum-style) lets a lightweight client fetch only the transactions that concern it. This is the single component that unlocks mobile wallets, hardware wallet support and exchange integrations alike — which is why it comes before all of them.

Objective	Status
Electrum-compatible indexing server	Planned
Public API endpoints for third-party services	Planned

Stage 3 — Mobile wallets

Built on Stage 2. Android first, since publication is faster and less constrained; iOS follows, subject to the App Store review process that applies to cryptocurrency applications.

Objective	Status
Android wallet	Planned
iOS wallet	Planned

Stage 4 — Hardware wallets

Krypton shares Litecoin's cryptographic foundations, so an existing hardware-wallet application can serve as the technical base. Publication, however, depends on the vendor's own review and approval process, which is outside the project's control — the objective is therefore stated as a submission, not as a guaranteed integration.

Objective	Status
Ledger application submission	Planned
Support for further hardware wallets	Planned

Stage 5 — Exchange availability

Listings are pursued progressively, from decentralised and smaller platforms towards larger ones. Major exchanges such as Binance or Kraken apply strict liquidity, volume and compliance requirements; they are a long-term ambition for the project rather than a scheduled deliverable, and no listing is announced before it is confirmed.

Objective	Status
Decentralised and smaller exchange availability	Planned
Mid-size exchange availability	Ambition
Major exchange availability (Binance, Kraken and similar)	Ambition

Stage 6 — Post-quantum research

A sufficiently powerful quantum computer would threaten the elliptic-curve signatures used by Krypton and every comparable blockchain. The exposure is not uniform: an address whose public key has never been revealed on-chain is protected behind a hash, which is far more resistant; the risk concentrates on addresses reused after spending from them.

Two things follow. First, an immediate and cost-free mitigation already exists — never reuse an address. The wallet generates a fresh address for every incoming payment, and the documentation will make this practice explicit. Second, adding post-quantum signatures would require a new output type and a consensus fork; standards are still maturing and no major chain has deployed them. This is therefore tracked as active research and monitoring, not as a dated deliverable.

Objective	Status
Address-reuse guidance and wallet defaults review	Planned
Monitoring of post-quantum signature standards	Research
Evaluation of a post-quantum output type	Research

12. Resources and disclaimer

Resource	Address
Source code	github.com/Kryptonproject-crypto/Krypton
Official website	krypton-explorer.org
Block explorer	explorer.krypton-explorer.org
Discord	discord.gg/g8c64BccQZ
X	x.com/Krypton_KYP

Disclaimer. This document is provided for informational and technical purposes only. It is not investment advice, a solicitation, or a promise of return. KYP is not a security and confers no rights over any legal entity. Cryptocurrencies are volatile and carry a risk of total loss; their regulatory status varies by jurisdiction. The software is distributed under the MIT licence, without warranty of any kind. Readers are encouraged to verify the code themselves and to commit only what they can afford to lose.